

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Bhuvaneshwar M J (C92511402) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.
3. A smart traffic management system is being designed to control traffic signals across a city based on vehicle density, pedestrian movement, and weather conditions. The system must respond to real-time sensor inputs while also learning traffic patterns over time to reduce congestion. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for minimizing congestion and travel time.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Smart home automation system is being designed to manage lighting, temperature and security.

Introduction:

A smart home automation system controls lighting, temperature and security using AI. It receives inputs from sensors such as motion detectors, temperature sensors and clocks, while learning user preferences to improve comfort, energy efficiency and safety.

Type of Intelligent Agent:

* Simple reflex Agent

⇒ Acts only on the current input using predefined rules.

⇒ Does not remember previous states.

Examples:

Turns on light when motion is detected

Turns off light when no motion is detected

* Model-Based Reflex Agent:

⇒ Maintains an internal model of the environment

⇒ Uses current and previous information to make decisions.

Examples:

Remembers that room is occupied recently.

Keeps the AC running even after the person leaves for few minutes.

* Goal Based Agent:

⇒ Make decisions to achieve specific goals

Examples:

Maintain room temp at 24°C

Lock all doors when everyone leaves the house

* Utility Based Agent

⇒ chooses the action that provides the highest overall benefit. Consider multiple factors simultaneously.

Best Approach:

A Hybrid Agent (Model Based + Goal Based + Utility-Based) is the most effective.

Reason:

- * Responds instantly to sensor inputs
- * learns user behaviour over time
- * Achieves comfort and security goals

Conclusion:

A Hybrid Agent provides the best solution because it combines quick responses, environmental awareness, planning and utility optimisation, ensuring comfort, energy efficiency and security.

Robot is placed in an unknown maze.

Introduction:

A Robot placed in an unknown maze must find the exit without knowing the map. A search algorithm help the robot explore paths until it reaches the goal.

Uniform Cost Search: (UCS)

Working:

- ⇒ Expand the node with the lowest cumulative path cost.
- ⇒ Guarantees the least-cost path.

Advantage

- ⇒ Finds the optimal solution
- ⇒ Suitable when path costs vary

Disadvantage:

- ⇒ High memory usage

Iterative Deepening Search (IDS)

Working:

- ⇒ Repeatedly performs Depth-First Search with increasing depth limits.
- ⇒ Combines DFS memory efficiency with BFS completeness.

Advantage:

- Very low memory usage
- Finds shallow solutions quickly

Disadvantage:

- ⇒ Repeats node expansions
- ⇒ Not optimal when path costs differ

Complexity:

Time: $O(b^d)$ Space: $O(bd)$

Relation to Intelligent Agents

(i) Simple Reflex agent

- * Moves based only on current surrounding
- * May get trapped in loops.

(ii) Model-Based Agent

- * Remembers visited locations
- Avoids revisiting the same paths.

(iii) Goal Based Agent:

Searches systematically to reach the maze exit

(iv) Utility Based Agent:

Chooses paths based on shortest distance

Combination:

Goal Based Agent with Uniform Cost Search is the most effective.

Justification:

- * The robot's goal is to reach the exit
- * UCS guarantees the least cost path
- * The goal based agent plans actions towards exit
- * Suitable for unknown environment, where path costs vary.

Conclusion:

For maze navigation a Goal Based Agent using uniform cost search provides the most reliable and optimal solution. These approaches enable efficient decision-making and successful navigation in unknown environments.